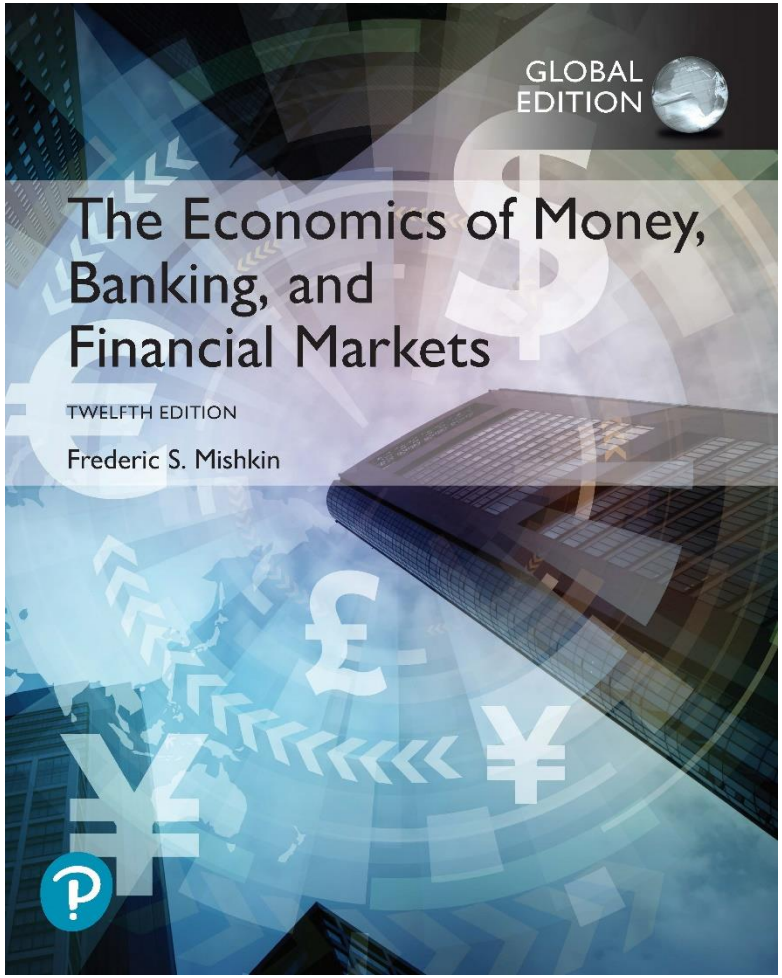
The Economics of Money, Banking, and Financial Markets

Twelfth Edition, Global Edition



Chapter 12

Financial Crises in Advanced Economies

Preview

- This chapter makes use of agency theory, the economic analysis of the effects of asymmetric information (adverse selection and moral hazard) on financial markets, to see why financial crises occur and why they have such devastating effects on the economy.
- It then applies our analysis to explain the course of events that led to a number of past financial crises, including the most recent global financial crisis.

Learning Objectives

- Define the term “financial crisis.”
- Identify the key features of the three stages of a financial crisis.
- Describe the causes and consequences of the global financial crisis of 2007–2009.
- Summarize the changes to financial regulation that developed in response to the global financial crisis of 2007–2009.
- Identify the gaps in current financial regulation and how they might be addressed with future regulatory changes.

Global: The European Sovereign Debt Crisis

- The increase in budget deficits that followed the financial crash of 2007–2009 led to fears of government defaults and a surge in interest rates.
- The sovereign debt debt, which began in Greece, moved on to Ireland, Portugal, Spain, and Italy.
- The stresses created by this and related events continue to threaten the viability of the euro.

What Is a Financial Crisis?

- A financial crisis occurs when there is a particularly large disruption to information flows in financial markets, with the result that financial frictions increase sharply and financial markets stop functioning.

Dynamics of Financial Crises

- Stage One: Initiation of a Financial Crisis
  - Credit Boom and Bust: Mismanagement of financial liberalization/innovation leading to asset price boom and bust
  - Asset-price Boom and Bust
  - Increase in Uncertainty
- Stage two: Banking Crisis
- Stage three: Debt Deflation

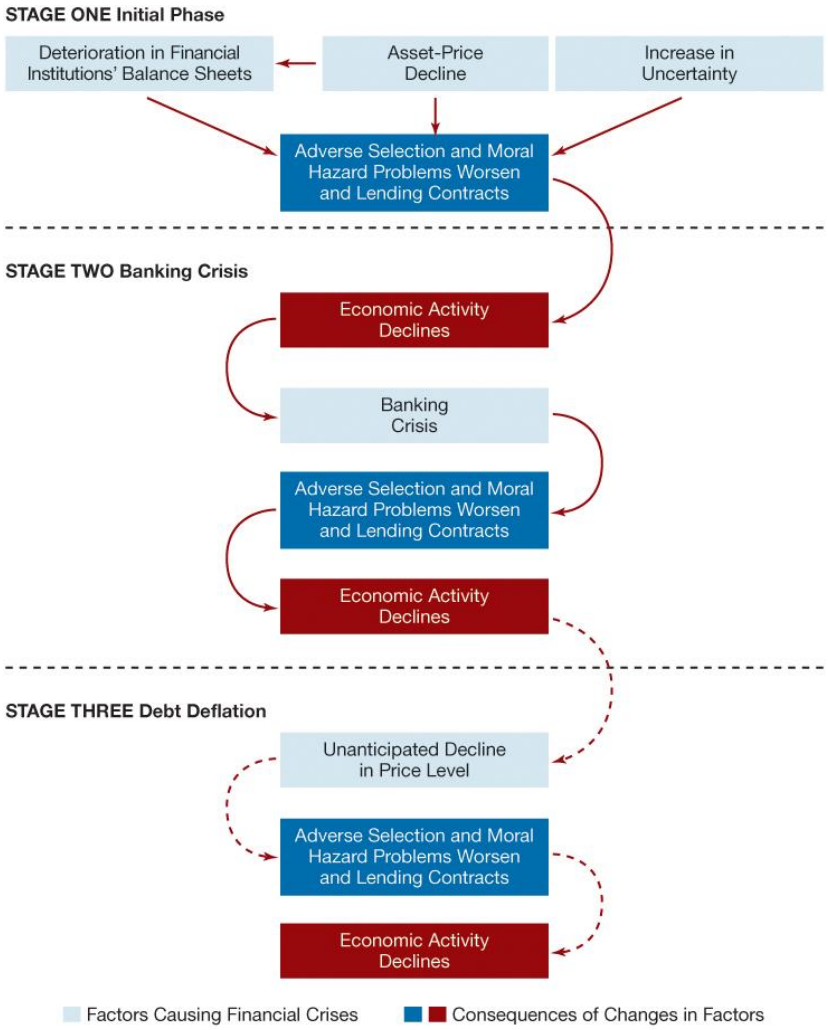
Dynamics of Financial Crises

CREDIT-MARKET SENTIMENT AND THE BUSINESS CYCLE*

DAVID LÓPEZ-SALIDO
JEREMY C. STEIN
EGON ZAKRAJŠEK

Using U.S. data from 1929 to 2015, we show that elevated credit-market sentiment in year $t-2$ is associated with a decline in economic activity in years t and $t+1$. Underlying this result is the existence of predictable mean reversion in credit-market conditions. When credit risk is aggressively priced, spreads subsequently widen. The timing of this widening is, in turn, closely tied to the onset of a contraction in economic activity. Exploring the mechanism, we find that buoyant credit-market sentiment in year $t-2$ also forecasts a change in the composition of external finance: net debt issuance falls in year t, while net equity issuance increases, consistent with the reversal in credit-market conditions leading to an inward shift in credit supply. Unlike much of the current literature on the role of financial frictions in macroeconomics, this article suggests that investor sentiment in credit markets can be an important driver of economic fluctuations. *JEL Codes*: E32, E44, G12.

Figure 1 Sequence of Events in Financial Crises in Advanced Economies



Bank panics: A frequent occurrence

Country	Number of banking crises since independence or 1800	Number of banking crises since 1945	Share of years in a banking crisis since independence or 1800
Argentina	9	4	8.8
Brazil	11	3	9.1
Canada	8	1	8.5
China	10	1	9.1
France	15	1	11.5
Germany	8	2	6.2
Japan	8	2	8.1
Russia	2	2	1.0
South Africa	6	2	6.3
United Kingdom	12	4	9.2
United States	13	2	13.0

Source: Adapted from Reinhart, Carmen M. and Kenneth Rogoff (2009), This Time is Different: Eight Centuries of Financial Folly, Princeton, NJ: Princeton University Press.

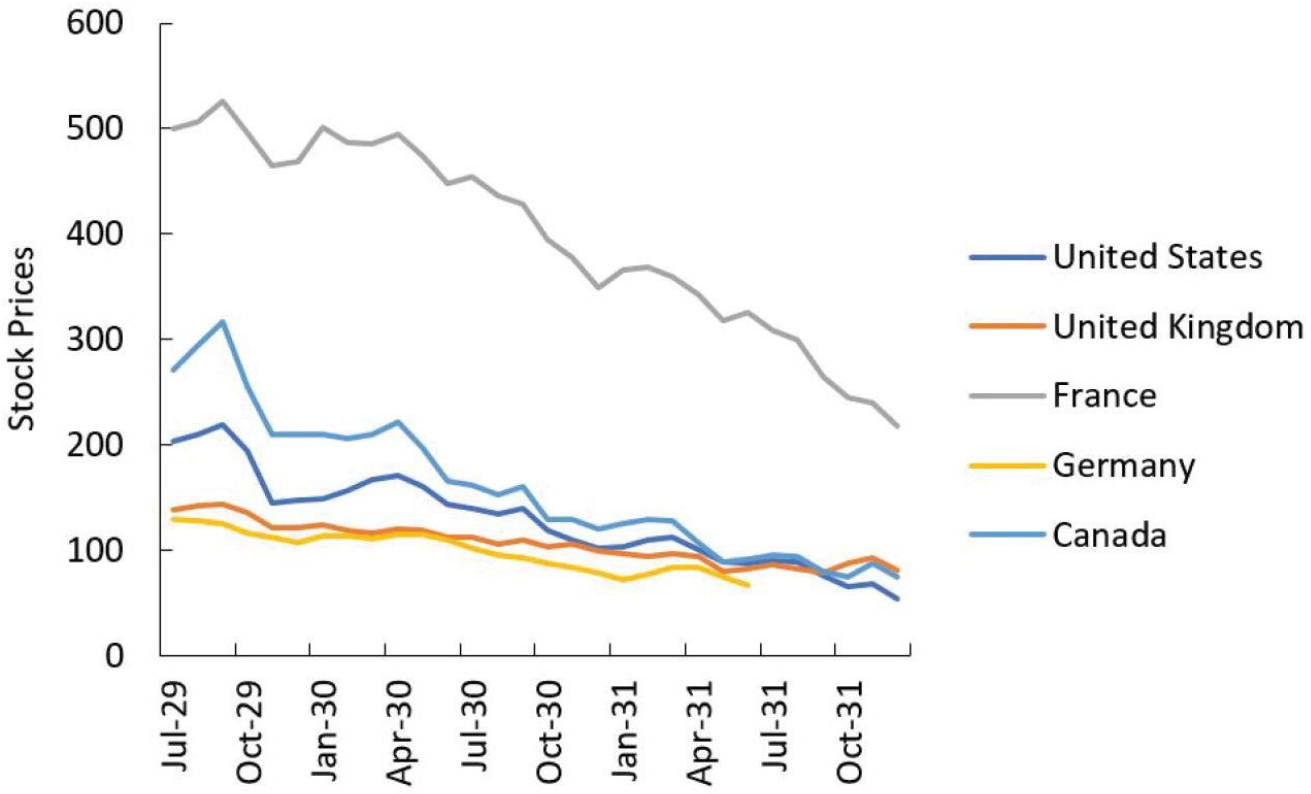
The Mother of All Financial Crises: The Great Depression

- How did a financial crisis unfold during the Great Depression and how it led to the worst economic downturn in the history of the United States and other advanced economies?
- This event was brought on by:
  - Stock market crash
  - Bank panics
  - Continuing decline in stock prices
  - Debt deflation

Figure 2 Stock Price Data During the Great Depression Period

Source: Dow-Jones Industrial Average (DJIA). Global Financial Data: http://www.globalfinancialdata.com/index_tabs.php?action=detailedinfo&id=1165 .

Figure 3 Stock Price Indexes in a Selected Number of Advanced Economies, 1929–1931



Source: Adapted from Kindleberger, 1986, Figure 6..

The Global Financial Crisis of 2007–2009

- Causes of the 2007–2009 Financial Crisis:
  - Financial innovations emerge in the mortgage markets
    - Subprime mortgage
    - Mortgage-backed securities
    - Collateralized debt obligations (CDOs)
  - Housing price bubble forms
    - Increase in liquidity from cash flows surging to the United States
    - Development of subprime mortgage market fueled housing demand and housing prices

The Global Financial Crisis of 2007–2009

- Causes (cont'd):
  - Agency problems arise
    - "Originate-to-distribute" model is subject to principal-(investor) agent (mortgage broker) problem
    - Borrowers had little incentive to disclose information about their ability to pay
    - Commercial and investment banks (as well as rating agencies) had weak incentives to assess the quality of securities
  - Information problems surface
  - Housing price bubble bursts

FYI Collateralized Debt Obligations and Credit Default Swaps (1 of 2)

- **Collateralized debt obligations** (CDOs) are credit derivatives that purport to redistribute the risks involved in lending.
- The creation of a CDO involves setting up a corporate entity called a *special purpose vehicle (SPV)* that buys a collection of assets such as corporate bonds and loans, commercial real estate bonds, and mortgage-backed securities.
- The SPV separates the payment streams (cash flows) from these assets into buckets that are referred to as tranches.
- The highest rated tranches, referred to as super senior tranches, are the ones that are paid off first and so have the least risk.

FYI Collateralized Debt Obligations and Credit Default Swaps (2 of 2)

- The lowest tranche of the CDO is the equity tranche and this is the first set of cash flows that are not paid out if the underlying assets go into default and stop making payments. This tranche has the highest risk and is often not traded.
- **Credit default swaps** are also credit derivatives. They are derivative contracts that allow the parties to exchange (swap) risks linked to cash flow payments
- A CDS seller commits himself/herself to compensate the buyer (lender) in the event of a default by a buyer (lender)'s client (borrower)

The Global Financial Crisis of 2007–2009

- Effects of the 2007–2009 Financial Crisis
  - After a sustained boom, housing prices in the U.S. and other countries (such as Ireland or Spain) began a long decline, starting in 2006.
  - The decline in housing prices contributed to a rise in defaults on mortgages and a deterioration in the balance sheet of financial institutions.
  - This development in turn caused a run on the shadow banking system.

The Global Financial Crisis of 2007–2009

- Origins: U.S. financial markets
- Crisis spreads globally
  - Sign of the globalization of financial markets
  - TED spread (3 months interest rate on Eurodollar minus 3 months Treasury bills interest rate) increased from 40 basis points to almost 240 in August 2007.

The Global Financial Crisis of 2007–2009

- Deterioration of financial institutions' balance sheets:
  - Write downs
  - Sell of assets and credit restriction
- High-profile financial firms fail, in the U.S. and Europe
  - In the United States: Bear Stearns (March 2008), Fannie Mae and Freddie Mac (July 2008), Lehman Brothers, Merrill Lynch, AIG, Reserve Primary Fund (mutual fund) and Washington Mutual (September 2008)

The Global Financial Crisis of 2007–2009

- High-profile financial firms fail, in the U.S. and Europe (contd.)
  - In the United Kingdom: Northern Rock (September 2008), Royal Bank of Scotland (October 2008)
  - Elsewhere: BPN (Portugal, November 2008), Anglo-Irish Bank (January 2009), the whole Icelandic banking sector (October 2008)

The Global Financial Crisis of 2007–2009

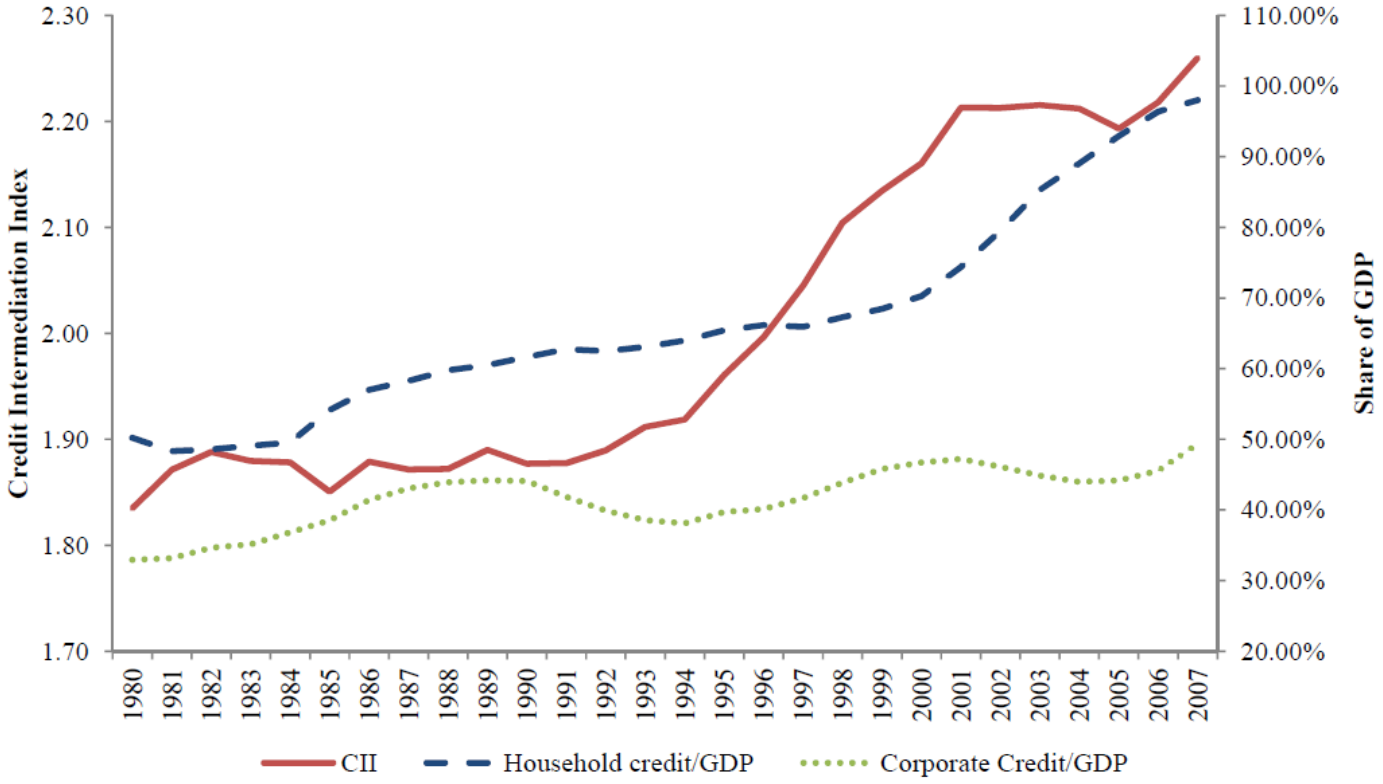
- Bailouts
  - In the United States, Congress approved a $787 billion economic stimulus plan on February 13, 2009.
  - In the United Kingdom, the government announced a £500 billion bank rescue package in October, 2008.
  - Similarly, in countries such as Ireland, Spain, Belgium, and the Netherland, large scale rescue plans were announced to bail out ailing banks.

The Global Financial Crisis of 2007–2009

Increased length of financial chain (Greenwood and Scharfstein (2013)**:**

Figure 11. Credit Intermediation Index

Notes: The Credit Intermediation Index is equal to the ratio of gross credit to net credit to end users (government, households, and nonfinancial firms). Household credit and corporate credit are from Table L1 of the Flow of Funds.

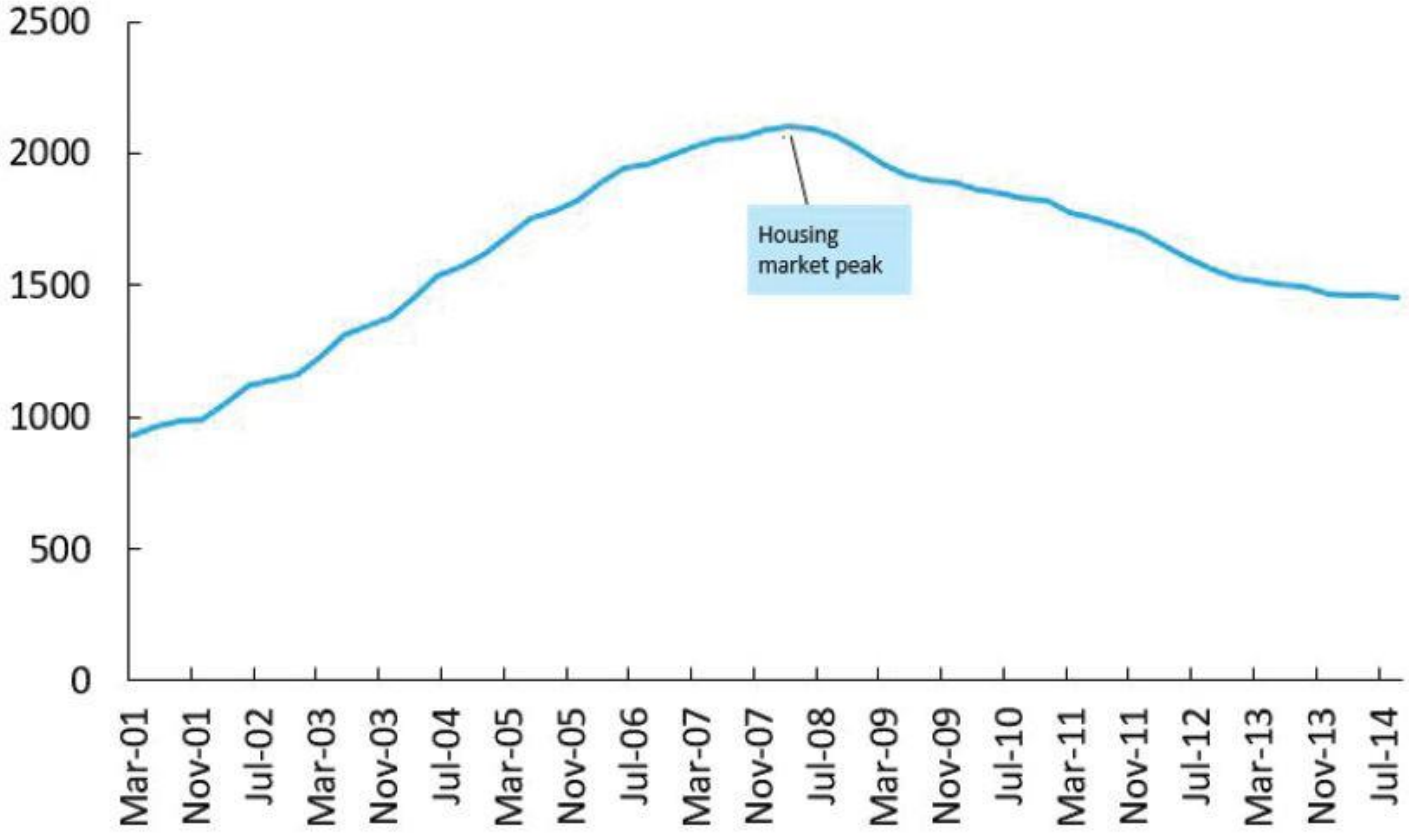


The Global Financial Crisis of 2007–2009

Procyclical lending (雨天收伞）

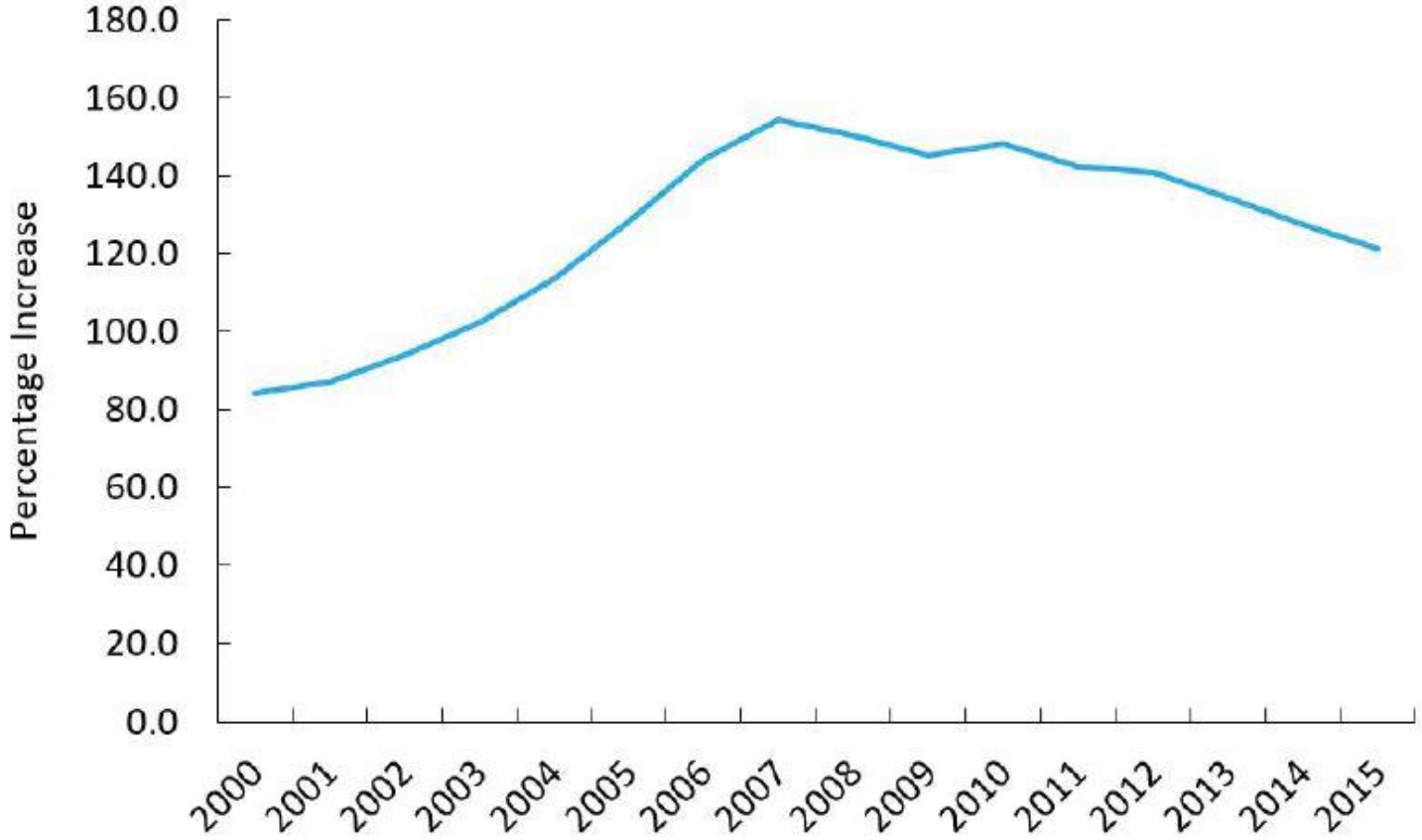
refers to supply-driven changes in lending that amplify the business cycle

Figure 4 Residential Housing Prices in Spain



Source: Bank of Spain, Financial Accounts.

Figure 5 Total Spanish Household Debt as a Percentage of National Income



Source: Bank of Spain, Financial Accounts..

Inside the Fed: Was the Fed to Blame for the Housing Price Bubble? (1 of 2)

- Some economists have argued that the low rate interest policies of the Federal Reserve in the 2003–2006 period caused the housing price bubble.
- Taylor argues that the low federal funds rate led to low mortgage rates that stimulated housing demand and encouraged the issuance of subprime mortgages, both of which led to rising housing prices and a bubble.

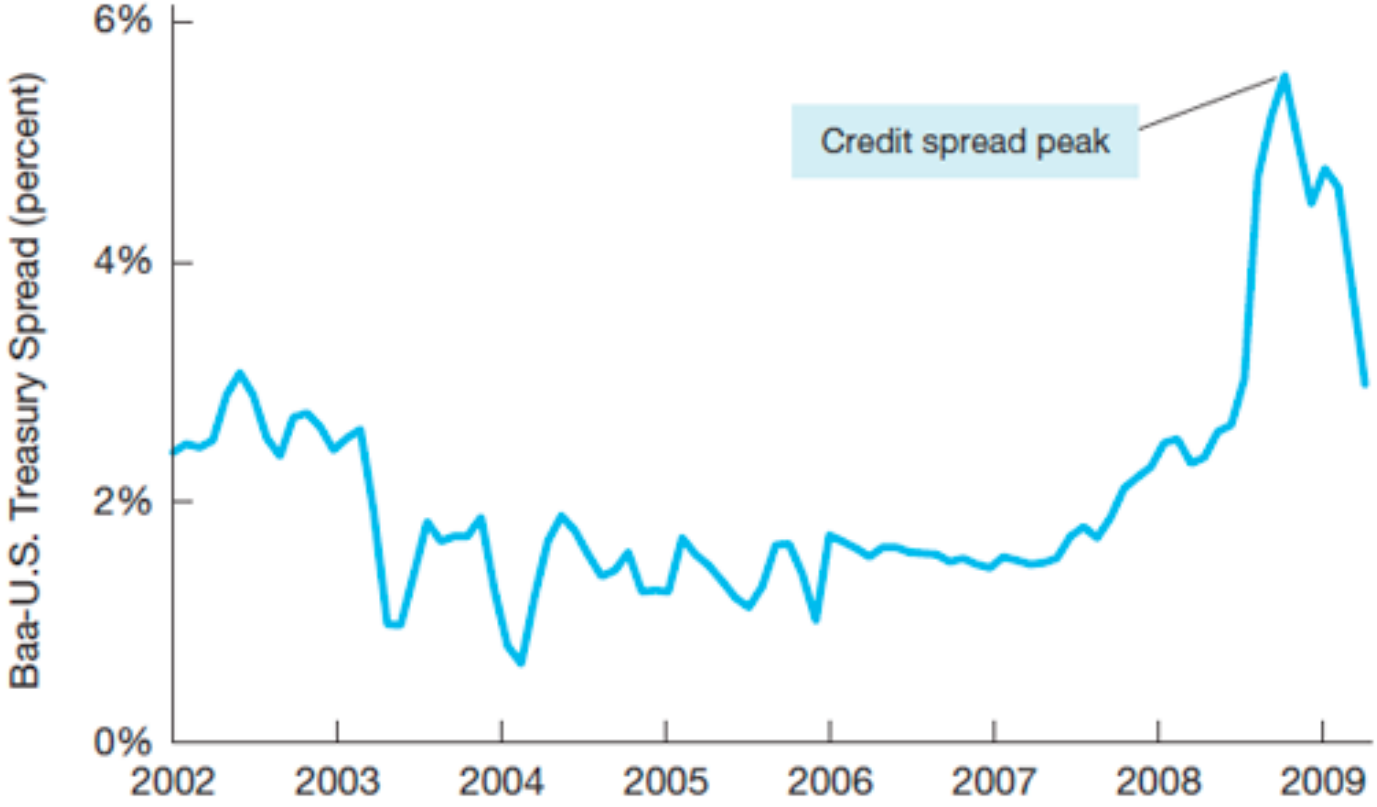
Inside the Fed: Was the Fed to Blame for the Housing Price Bubble? (2 of 2)

- Federal Reserve Chairman Ben Bernanke countered this argument saying the culprits were the proliferation of new mortgage products that lowered mortgage payments, a relaxation of lending standards that brought more buyers into the housing market, and capital inflows from emerging market countries.
- The debate over whether monetary policy was to blame for the housing price bubble continues to this day.

The Global Financial Crisis of 2007–2009

- Height of the 2007–2009 Financial Crisis
  - The stock market crash gathered pace in the fall of 2008, with the week beginning October 6, 2008, showing the worst weekly decline in U.S. history.
  - Surging interest rates faced by borrowers led to sharp declines in consumer spending and investment.
  - The unemployment rate shot up, going over the 10% level in late 2009 in the midst of the “Great Recession, the worst economic contraction in the United States since World War II.
- Political consequences: electoral defeat of incumbent governments (Iceland, UK), unrest, increased polarization

Figure 6 Credit Spreads and the 2007–2009 Financial Crisis



Source: Dow-Jones Industrial Average (DJIA). Global Financial Data: http://www.globalfinancialdata.com/index_tabs.php?action=detailedinfo&id=1165.

Government Intervention and the Recovery

(1 of 2)

- Short-term Responses and Recovery
  - Financial Bailouts: In order to save their financial sectors and to avoid contagion, financial support was provided by many governments to bail out banks, other financial institutions, and even the so-called “too-big-to-fail” firms that were severely affected by the financial crisis.
  - Fiscal Stimulus Spending: To boost their individual economies, most governments used fiscal stimulus packages that combined government expenditure and tax cuts.

Government Intervention and the Recovery (*contd.*) (2 of 2)

- Short-term Responses and Recovery
  - Japan's consecutive stimulus packages, totaling $568 billion, were among the highest during the crisis, but these proved largely ineffective
  - European nations showed moderate success.

Global: Latvia's Different and Controversial Response (1 of 2)

- Latvia's independence from the USSR in 1991 and its fiscal policies helped it join the EU, 2004; and the Eurozone, 2014.
- Latvia's economic policies had a low budget deficit and a fixed exchange rate against the Euro.
- In 2007, the country's second-largest bank, Parex Bank, collapsed. Latvia needed €7.5 billion to recapitalize and meet external financing requirements.

Global: Latvia's Different and Controversial Response (2 of 2)

- Austerity program: citizens voluntarily endured the layoff of 25% of state workers, 40% salary cuts, and social expenditure reductions.
- After a contraction of over 25%, the country's GDP started to grow to its near pre-crisis levels.
- Expansionary Contraction: Success in Latvia, but maybe inapplicable in other countries because of the political motivations involved.

Stabilizing the Global Financial System: Long-Term Responses

- With the individual emergency national bailouts to rescue national economies and financial sectors, global leaders looked to building a more stable and robust global financial system. Steps taken by governments included
  - Implement sound macroeconomic policies
  - Enhance their financial infrastructure
  - Develop financial education and consumer protection rules
  - Enact macro- and microprudential regulations.

Long-Term Responses (contd.)

- At the international level
  - proactive globally-binding supervision was designed
  - financial market discipline enforced
  - systemic risk managed
- To avoid collective action problems and to ensure that policy actions are mutually consistent with national growth objectives, aggregate plans began to be drafted simultaneously. The first ever of these is the Mutual Assessment Process (MAP) launched in 2009 by the G20.

FYI The LIBOR Scandal (1 of 2)

- LIBOR stands for “London Interbank Offered Rate”; it is an indicator of the cost of short-term money on the inter-bank market, calculated daily by Thomson-Reuters on the basis of reports by a sample of large banks.
- It is the world's most important benchmark interest rates for hundreds of trillion dollars worth of financial products around the world.

FYI The LIBOR Scandal (2 of 2)

- In June 2012, it was revealed that traders at four large banks had colluded since 2005 to manipulate the LIBOR, so as to make their banks appear to be in much healthier financial situation than they were in fact.
- Given the financial significance of LIBOR, the scam was enormous.
- After lengthy investigations, the four banks were fined $6 billion and criminal charges were pressed against those directly responsible.

Future Regulations and Policy Areas at the International Level (1 of 3)

- The gravity of the 2007-08 crisis requires a more radical overhaul of the financial supervisory and regulatory system at the international level.
- Such overhaul would also help decrease the risks of **regulatory arbitrage**, through which financial actors take advantage of loopholes in national regulatory systems.
- Steps have recently been taken in this direction by regulators around the world, in four areas

Future Regulations and Policy Areas at the International Level (2 of 3)

- Bilateral and multilateral supervisory coordination
  - Several initiatives of cross-country supervisory coordination have been adopted since the crisis, to help deal with liquidity shortages and global risk profiles
  - However, such initiatives are of limited effect, given their voluntary, non-binding nature
- Collective supervisory cooperation
  - The G20 set up the **Financial Stability Board** in 2009
  - Aims to reveal regulatory gaps & harmonize financial regulation

Future Regulations and Policy Areas at the International Level (3 of 3)

- Collectively coordinated macroeconomic plans
  - Needed to contribute to financial stability
  - **Mutual Assessment Plan** launched by the G20 in 2009
- Self-discipline
  - Required to avoid the irresponsible and unethical behavior of bankers in the years leading to the 2007-08 crisis.
  - Various initiatives, such as the 2012 establishment of the Systemic Risk Council in the United States.